MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1002.3
STANDARD	**02 March 2023**
Range Motion Imagery	

1 Scope

This standard describes Range Motion Imagery, its format and supporting metadata. Range Motion Imagery is a temporal sequence of Range Images. Each Range Image is a collection of Range Measurements from a sensor to target scene. A Range Measurement is the distance (e.g., meters) from an object (or area) in the scene to the sensor. This standard uses a KLV structure to store Range Images, which systems can multiplex into MISB-conformant Motion Imagery streams and files. The KLV structures provide flexible, efficient packing, and future extensibility. Systems can produce Range Motion Imagery either standalone or in collaboration with other Motion Imagery.

This standard describes: Perspective Range Motion Imagery and Depth Range Motion Imagery; the collection methods of Range Motion Imagery; the formats used for storing or transmitting Range Motion Imagery; the supporting metadata needed for Range Motion Imagery including, temporal, uncertainty, and compression parameters; and the alignment to Collaborative Imagery.

2 References

MISB references cited here-in reflect versions current to the publication date of a document. In the event of a MISB document correction, the corrected document will have a single letter Minor Version appended to the Major Version number per the MISB Document Development Process [1]. For example, corrections to ST 0001.2, which has a Major Version of 2, result in a new version titled ST 0001.2a, which includes a Minor Version of "a". The MISB will not update the referring document (this document) with the Minor Version number change. When acquiring any MISB reference listed below from an NGA repository the latest version may be either a Major or Minor Version.

[1] MISB Document Development Process 7.1, Dec 2021.
[2] MISB ST 1202.3 Generalized Transformation Parameters, Mar 2023.
[3] MISB ST 0807.26 MISB KLV Metadata Registry, Jun 2022.
[4] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.
[5] MISB MISP-2023.2: Motion Imagery Handbook, Mar 2023.
[6] MISB ST 0603.5 MISP Time System and Timestamps, Oct 2017.
[7] MISB ST 1303.2 Multi-Dimensional Array Pack, Jun 2020.
[8] MISB ST 1201.5 Floating Point to Integer Mapping, Jun 2021.

3 Revision History

Revision	Date	Summary of Changes
ST 1002.3	03/02/2023	• Retired requirements 04, 10, 11, 12 • Added requirements 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 • Changed Item 11 from ber-oid to uint • Changed Item 12 from ber-oid to uint8 • Changed MDARRAY to MDAP in Section Data VLP (Table 4) • Removed Universal Label's (keys) from tables and changed "element" to "item" for LS items consistent with current MISB practice • Updated Range Image Local Set Table formats to match ST 0107 and MI Handbook, consistent with current MISB practice • Editorial changes to improve readability • Updated references

4 Acronyms and Definitions

KLV	Key-Length-Value
LS	Local Set
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
SACP	Single Aim Center Pixel
SPRM	Single Point Range Measurement
ST	Standard
VLP	Variable Length Pack

5 Introduction

Range Motion Imagery is a series of Images where the value of the Range Image Pixels is the distance from the Sensor to a Scene Area. This standard provides for two types of Range Imagery and multiple sources of Range Imagery.

5.1 Range Imagery Types

Range Imagery data values can represent two different types of measurements: **Perspective Range Measurements** or **Depth Range Measurements**. **Perspective Range Measurements** represent measured or computed distances from the Scene Area to the perspective center of the sensor, as shown in Figure 1. Perspective Range data directly integrates with a rigorous sensor model.

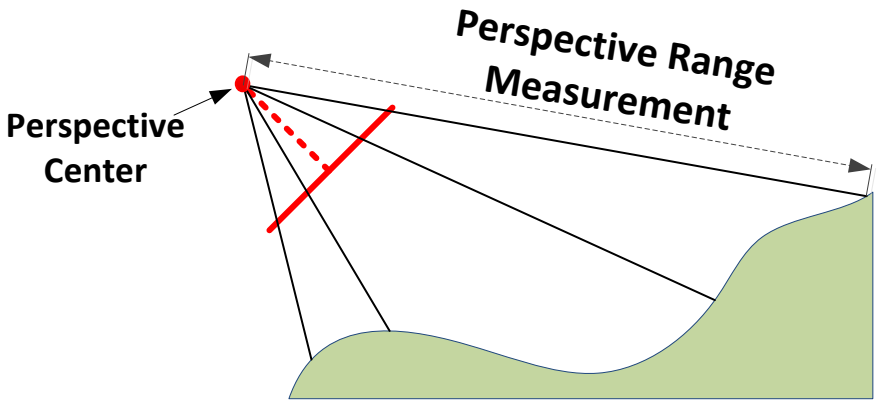


Figure 1: Illustration of Range Measurements (profile view)

Depth Range Measurements represent measured or computed distances from the Scene Area to a plane parallel to the focal plane, as shown in Figure 2. In this standard, the plane parallel to the focal plane is the Backplane.

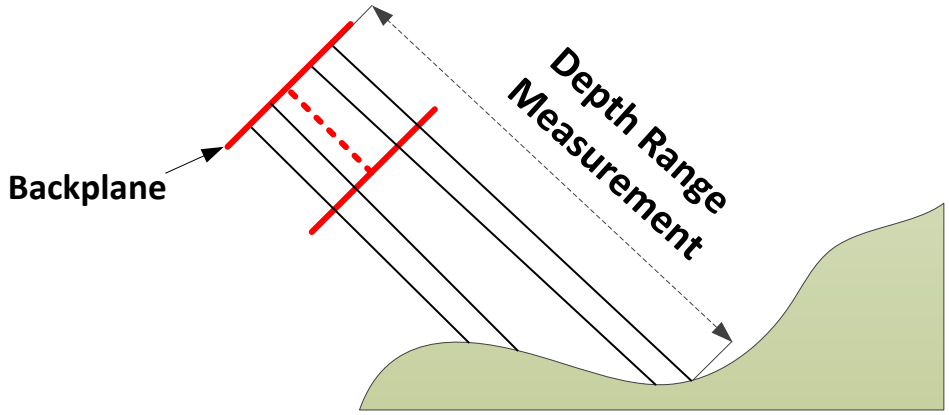


Figure 2: Illustration of Depth Range Measurements (profile view)

Depth data are more commonly known as depth fields or depth maps. Depth data does not integrate into rigorous sensor models; however, systems can convert depth data to range data. This standard does not provide such conversion algorithms.

5.2 Range Imagery Sources

The source of Range Imagery may come directly from a **Range Sensor** or **Computationally Extracted** from another imagery source (e.g., Camera) or data source (e.g., Point Cloud).

5.2.1 Range Sensor

When collecting Range Imagery directly from a **Range Sensor,** the range data is either stand-alone or relates to another Sensor, known as a Collaborative Sensor, e.g., visible light and infrared (IR) cameras. With a Collaborative Sensor, the Range Imagery is either co-boresighted or non-boresighted with the Collaborative Sensor. In co-boresighted Range Imagery, the sensor

collects data through the same physical aperture as a Collaborative Sensor, which provides identical geometry to the Collaborative Sensor's collected image, as Figure 3 illustrates.

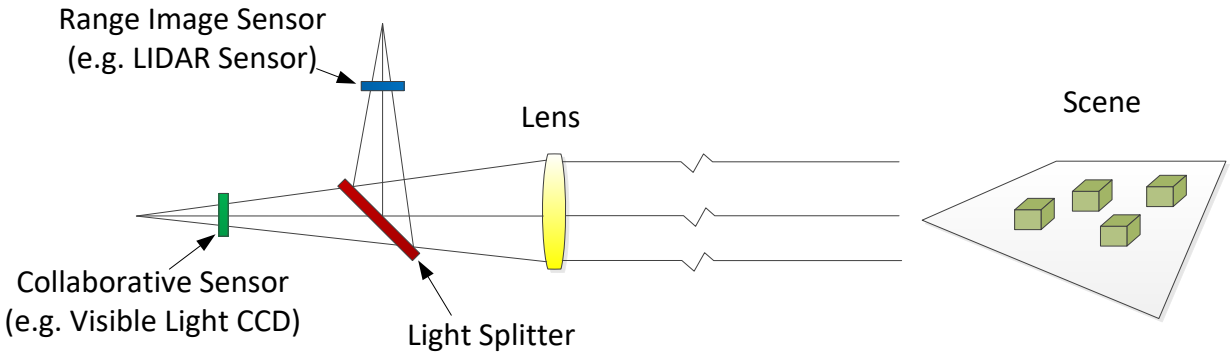


Figure 3: Co-Boresighted System of Sensors

Although the geometry is the same, Range and Collaborative Images may not have the same orientation, pixel sizes or magnification through the optical path. The Generalized Transformation Local Set (LS) (i.e., MISB ST 1202 [2]) enables systems to provide the mathematical parameters for mapping the data from one image perspective to the other. Systems may include the Generalized Transformation LS with the Range data for later processing.

When the Range Imagery does not meet the requirements for being co-boresighted, the Range Imagery is non-boresighted. Figure 4 illustrates a non-boresighted system of Sensors.

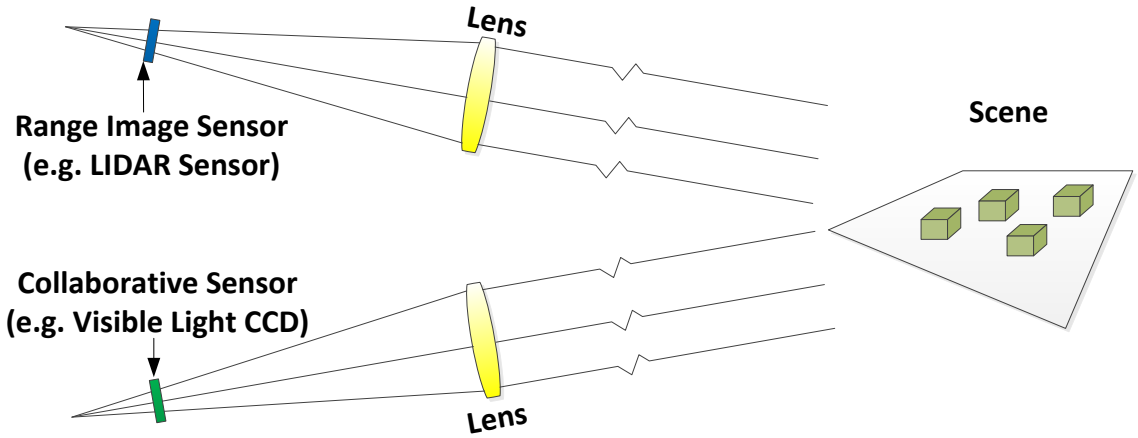


Figure 4: Non-Boresighted System of Sensors

With non-boresighted imagery, systems may need to align the Range Imagery with the Collaborative Imagery using an image perspective transformation. The Generalized Transformation LS enables systems to provide the mathematical parameters for perspective mapping the data from one image perspective to the other.

5.2.1.1 Single Point Range Measurement

In Range Imagery collected using a Collaborative Sensor, the Range Image dimensions (rows and columns) may be different than the Collaborative Sensor. A Range Image can be under-

sampled or over-sampled as compared to the Collaborative Sensor's Image. The minimum size of the Range Image can be as small as one range measurement; this is a special case of Range Image data called a Single Point Range Measurement (SPRM).

A SPRM is associated with a Collaborative Sensor's Image. The SPRM is a measurement from either the perspective center, or Backplane of the Collaborative Sensor, through the image plane to a point on the scene. Because the SPRM may not be in the exact center of the Collaborative Sensor's image, the row and column location of the image intersection pixel contributes to this measurement. When the SPRM passes through the center pixel of the image, the range measurement is a Single Aim Center Pixel Range Measurement, or SACP Range Measurement. Figure 5 and Figure 6 illustrate a SPRM and a SACP Range Measurement, respectively. The illustrations show the SPRMs as Perspective Range Measurements; however, the SPRMs can also be Depth Range Measurements.

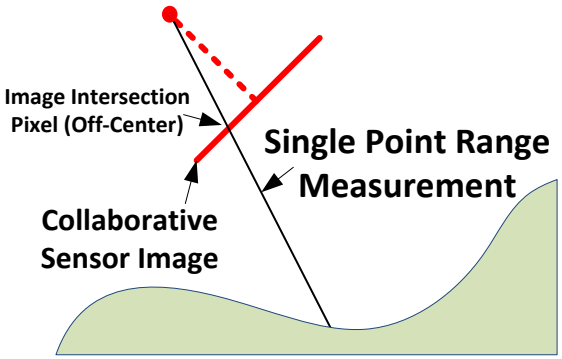


Figure 5: SPRM

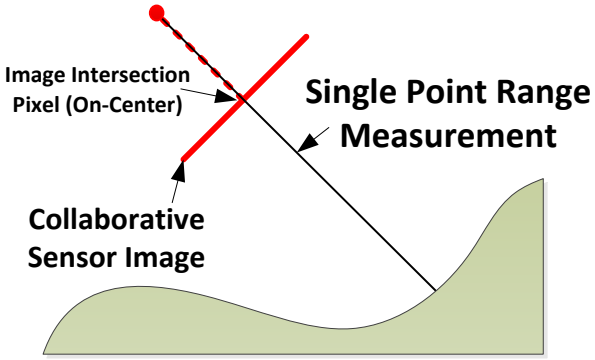


Figure 6: SACP Range Measurement

5.2.2 Computationally Extracted

Computationally Extracted Range Imagery infers the range data from optical flow, or other image/data processing techniques as an alternative to active sensor measuring.

5.3 Range Imagery Data Uncertainty

Since Range Imagery is a set of measured or computed distances, there are errors associated with these values. To maximize the utility of the Perspective Range or Depth Range data, this document includes the formatting method for the uncertainty of the Range Imagery at each pixel in the range image. The sensor developer computes, estimates, or measures (or a combination of the three) the uncertainty. The uncertainty is a system's ability to measure range reliably and varies for each ranging system. The Range Measurement uncertainty is a standard deviation (sigma, σ_i) in meters along the Range Measurement vector.

6 Data Representations

This standard supports two data representations: **Range Image** and **SPRM**. A **Range Image** representation consists of a two-dimensional array of data, the range data's uncertainty,

transformation information, and support values. A **SPRM** representation consists of a single range value, the range value's uncertainty, row, and column locations.

This standard defines the Range Image LS which supports both data representations. Table 1 lists the Range Image LS categories of items which support the two different data representations.

Table 1: Range Image LS Categories

Category	Description
Administration	Required administrative information (i.e., Version number and CRC)
Support	Required support information for describing the Range Image (e.g., image time, Range Imagery type)
Single Point Range	Information for describing and supporting the SPRM data representation
Range Image	Information for describing and supporting the Range Image data representation

Range Motion Imagery is a series of Range Images (or SPRMs) with information for each Range Image described by a Range Image LS.

Table 2 lists the metadata items for the Range Image LS, which has the following columns:

- **Tag** is the LS item's tag number
- **Name** is the name of the item
- **Category** indicates the purpose of the data item
- **Section** is the section number (in this document) providing the full description for the item
- **Format** is the data type for the item following type names in ST 0107 and Motion Imagery Handbook, e.g., vlp = Variable Length Pack
- **M/O** denotes the LS item is either Mandatory (M) or Optional (O)

Table 2: Range Image Local Set

Tag	Name	Category	Section	Format	M/O
1	Range Image Precision Time Stamp	Support	6.2.1	uint64	M
2-10	**Retired (Do Not Use)[1]**				
11	Document Version	Administration	6.1.1	uint	M
12	Range Image Enumerations	Support	6.2.2	uint8	M
13	SPRM	Single Point Range	6.3.1	float	O
14	SPRM Uncertainty	Single Point Range	6.3.2	float	O
15	SPRM Row Coordinate	Single Point Range	6.3.3	float	O
16	SPRM Column Coordinate	Single Point Range	6.3.3	float	O
17	Number of Sections in X	Range Image	6.4.2	ber-oid	O
18	Number of Sections in Y	Range Image	6.4.2	ber-oid	O
19	Generalized Transformation LS	Range Image	6.4.3	ls	O
20	Section Data Variable Length Pack	Range Image	6.4.1	vlp	O
21	CRC-16-CCITT	Administration	6.1.2	uint16	M
22	Leap Seconds	Administration	6.2.2	int	O
51-54	**Retired (Do Not Use)[1]**				

[1]Retired items from previous versions of this document.

The Universal Label (Key) for the Range Image LS is (see MISB ST 0807 [3]):

06.0E.2B.34.02.0B.01.01.0E.01.03.03.0C.00.00.00 (CRC 41152)

MISB ST 0107 [4] provides a set of baseline requirements for applying KLV metadata in Motion Imagery.

Requirement	
ST 1002.1-02	All Range Image Local Set metadata shall be expressed in accordance with MISB ST 0107.

6.1 Administrative Information

There are two Administrative Information items in the Range Image LS: Version Number and CRC Error Detection.

6.1.1 Item 11: Version Number

The Version Number of the LS is the same as the minor version number of this document. For example, with this version of MISB ST 1002.3, the version number value is ‘3’.

Requirement	
ST 1002.1-03	A Range Image Local Set shall include a Version Number.

A Range Imager LS parser should validate the version number to ensure the parser is properly parsing the LS, and the Section Data Variable Length Pack (defined in Section 6.4.1).

6.1.2 Item 21: CRC Error Detection

The last item in the Range Image LS is a two-byte CRC (CRC-16-CCITT) which aids in detecting transmission errors. When embedding the Range Image LS in a parent LS this item is not necessary. For standalone use, per instructions in the Motion Imagery Handbook [5], compute the CRC across the entire LS packet starting with the 16-byte LS key and ending with the length field of the CRC data item.

When receiving a Range Image LS, if the CRC computation results do not match the CRC stored in the packet, discard the packet as being invalid.

Requirement(s)	
ST 1002.3-24	Where the Range Image LS is standalone, the Range Image LS shall include the CRC-16-CCITT (Item 21) in accordance with the Motion Imagery Handbook.
ST 1002.3-25	Where the Range Image LS is embedded in a parent LS, the CRC-16-CCITT (Item 21) shall be omitted.

Refer to Motion Imagery Handbook, "CRC Algorithms" Appendix, for details and sample code for computing the CRC.

6.2 Range Motion Imagery Support Data

Range Motion Imagery requires support data to properly interpret the Range Imagery. The Range Image LS includes three support data items: Range Image Precision Time Stamp, Leap Seconds, and Range Encoding Enumerations.

6.2.1 Item 1: Range Image Precision Time Stamp

The Range Image Precision Time Stamp is the time when the measurements of the Range Image occurred. This time information enables coordination between the Range Image and other sources of data, such as a Collaborative Sensors image or other sensor data. The time value is an invocation of the MISP Precision Time Stamp, a 64-bit unsigned integer representing the number of microseconds since midnight of January 1st 1970 (1970-01-01T00:00:00Z) without leap-seconds, as defined in MISB ST 0603 [6] and detailed in the Motion Imagery Handbook. To convert the Precision Time Stamp to UTC, use the algorithm in the Motion Imagery Handbook. For the UTC conversion, the number of leap seconds may come from either the KLV packet (Item 22 - Leap Seconds), or from a current published leap second table.

Requirement(s)	
ST 1002.1-05	A Range Image Local Set shall include a Precision Time Stamp (Item 1) as defined in MISB ST 0603.
ST 1002.1-06	The Precision Time Stamp (Item 1) shall appear as the first metadata item within a Range Image Local Set.

Positioning the Precision Time Stamp as the first item facilitates rapidly checking whether the LS matches the desired time for processing a collaborative image.

6.2.2 Item 22: Leap Seconds

Leap Seconds provides the positive or negative offset between the Precision Time Stamp (Item 1) and UTC; see the Motion Imagery Handbook for more details on the MISP Time System, UTC Time System, Leap Seconds, and conversions between MISP and UTC Time Systems.

When adjusting a Precision Time Stamp to UTC multiply the leap second value by 1,000,000 to convert it to microseconds.

6.2.3 Item 12: Range Image Enumerations

Range Image Enumerations is an integer value containing three enumerated values as shown in Table 3. The Range Image Enumeration value combines the **Range Image Source**, **Range Imagery Data Type**, and the **Range Imagery Compression Method** enumerations into a single byte. The most significant bit (bit 7) is a reserved (Res.) value and always set to zero. The remaining least significant seven bits (6 – 0) decode into the three separate enumerations.

Table 3: Range Image Enumerations

MSB ← Bits → LSB							
7	6	5	4	3	2	1	0
Res.	**Range Image Source**	**Range Imagery Data Type**			**Range Imagery Compression Method**		
0	0 = Computationally Extracted 1 = Range Sensor	000 = Perspective Range Image 001 = Depth Range Image 010 - 111 = Reserved			000 = No Compression 001 = Planar Fit 010 - 111 = Reserved		

Range Image Source declares Range Imagery created either from a Range Sensor or Computationally Extracted, as Section 5 describes. This enumeration has only two values, thus consuming one bit of the Range Image Enumerations value.

Range Imagery Data Type declares the type of Range Imagery, either Perspective Range Image or Depth Range Image, as Section 5 describes. Currently, there are two defined values: 0=Perspective Range Image and 1=Depth Range Image. A reserved space allows for defining additional types in the future.

Range Imagery Compression Method declares the encoding systems method of compression of the Range Image. There are currently only two methods: No Compression and Planar Fit. The *No Compression* method means data is in its raw form. Section 7 describes the *Planar Fit* method. A reserved space allows for defining additional compression types in the future.

Although the Range Image Enumerations value is currently one byte, it is possible this value could grow to additional bytes in future versions of this standard. Future additions will only add more significant bytes and bits; therefore, the least significant byte format will not change as described in Table 3.

6.3 Single Point Range Measurements

A SPRM is one data representation the Range Image LS supports. The SPRM data representation includes: SPRM value, SPRM Uncertainty, and SPRM Row and Column Coordinates.

6.3.1 Item 13: SPRM

The SPRM value is the measure of distance (in meters) from either the principal point, or Backplane of a Collaborative Sensor through the image plane to a point in the scene; see 5.2.1.1. This value is either a 32-bit or 64-bit IEEE floating point value.

6.3.2 Item 14: SPRM Uncertainty

The SPRM Uncertainty is the uncertainty (sigma, σ) of the SPRM data, in meters, along the measured vector from either the perspective center or Backplane to the scene. This value is either a 32-bit or 64-bit IEEE floating point value. The implementing system determines the uncertainty; see Section 5.3.

6.3.3 Item 15 & 16: SPRM Row and Column Coordinates

The SPRM Row and Column Coordinates is the location within the Collaborative Image the SPRM passes through. For example, Figure 5 illustrates the SPRM passing through the Collaborative Sensor Image, where the coordinates of the Image Intersection Pixel are the SPRM Row and Column Coordinates values.

The row and column values are either 32-bit or 64-bit IEEE floating point values. When omitting the Row and Column values from the Range Image LS, the default values are set to the center of the Collaborative Sensor's Image.

Requirement(s)	
ST 1002.1-07	Where a Range Image LS does not include a Single Point Range Measurement Row Coordinate (Item 17), the default value shall be set to the center Row of the image.
ST 1002.1-08	Where a Range Image LS does not include a Single Point Range Measurement Column Coordinate (Item 18), the default value shall be set to the center Column of the image.

6.4 Range Image

A Range Image is one data representation the Range Imagery LS supports. The Range Image data representation includes Range Measurements, Uncertainty values, and Generalized Transformation data. The Range Measurements are a rectangular array of distances (in meters) from either the principal point, or Backplane, to points in the scene (see Section 5.2.1.1). The Uncertainty values are a rectangular array with the same dimensions as the Range Measurements array. Each Range Measurement value has an uncertainty value. If the uncertainty values are not known, the uncertainty array size is set to zero as a signal of "unknown uncertainty".

To facilitate an improvement in data compression, this standard specifies the partitioning of the Range Measurement array into separate parts, where each part is a "Section". Sections are rectangular areas that when combined form the full Range Measurement array. Systems may choose to format the whole Range Measurement array in one section.

Compressing each Section provides efficient transmission and storage. Sections have two different layouts, **simple** or **complex**. A **simple** Section layout divides the image into either

horizontal or vertical strips. All horizontal strips have the same width as the full image but can vary in height as illustrated in Figure 7. All vertical strips have the same height as the full image but can vary in width as illustrated in Figure 8.

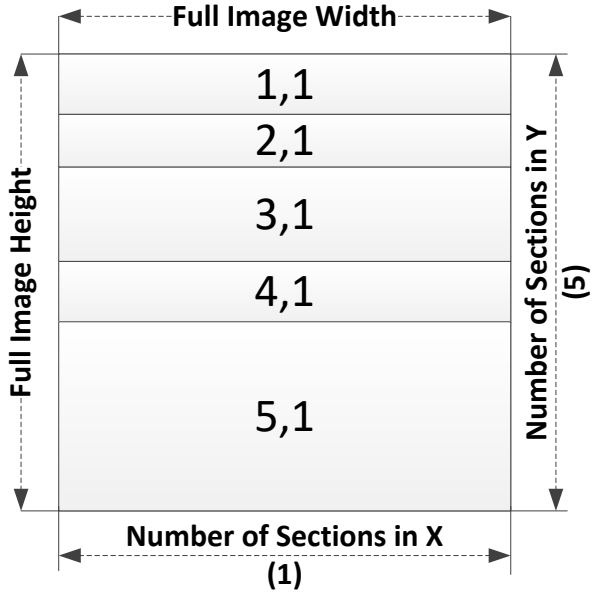


Figure 7: Illustration of five Horizontal Sections

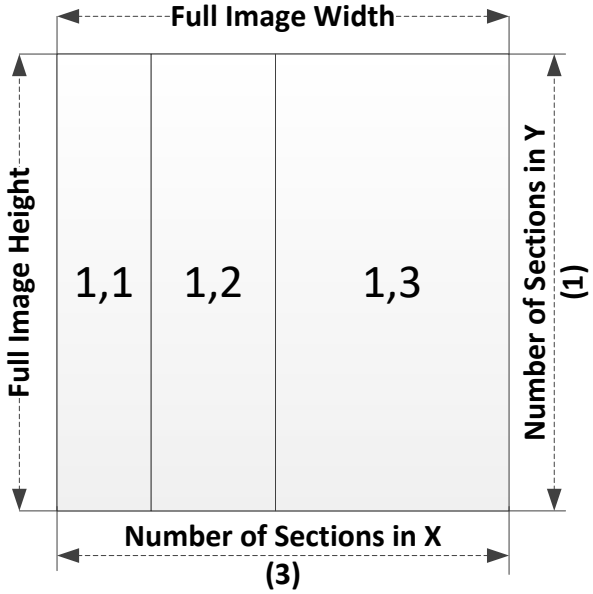


Figure 8: Illustration of three Vertical Sections

A **complex** Section layout divides the image into various size rectangles where each rectangle does not necessarily span the full image width or height as illustrated in Figure 9.

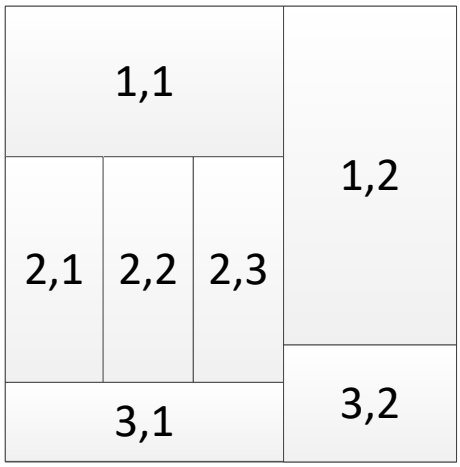


Figure 9: Illustration of a Complex Section layout

Currently, this standard only allows simple Section layout; however, future versions of this standard may allow complex Section layout.

Requirement	
ST 1002.1-09	Where formatting a Range Image into separate sections, the section layout shall only use the Simple Section layout method.

A Section Data Variable Length Pack (VLP) (Item 20) construct carries the data for one Section of the Range Measurement array. Thus, the Range Image LS may contain multiple Item 20's within the LS. Specifically, the number of Section Data VLPs equals the Number of Sections in X (Item 17) multiplied by the Number of Sections in Y (Item 18).

Each Section Data VLP will also include: the coordinates of the Section (in the overall Range Measurement array), the uncertainty values for the Section data, and optional compression parameters.

6.4.1 Item 20: Section Data VLP

The information in each Section Data VLP includes section coordinates, section range values, section uncertainty values, and optional compression parameters. Figure 10 illustrates a Section Data VLP with each item in the VLP prefixed with its item length. The green items indicate required values (along with their lengths in blue); the pink item can be "zeroized" (see Section 6.4.1.3); and the yellow items optionally truncated (along with their lengths). As a group, systems may truncate the optional compression parameters (yellow) from the VLP (along with their lengths); that is, either the VLP includes all three compression parameters or none of the three compression parameters.

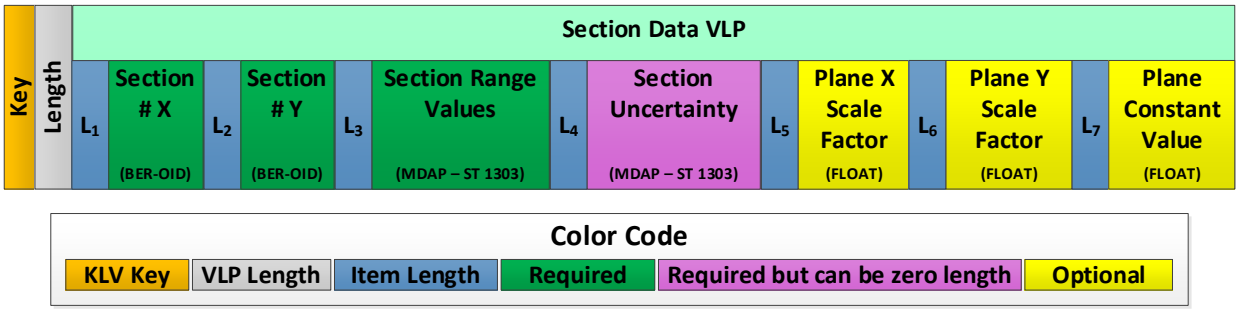


Figure 10: Illustration of Section Data Variable Length Pack

Table 4 lists the Section Data VLP elements in order. Table 4 includes the following columns:

- **Order** is the position the element is in the VLP
- **Name** is the name of the pack element
- **Notes** provide additional information about the element
- **Section** is this document's section number that provides details on the VLP element
- **Format** is the data type used for the value
- **MZO** indicates if an element is Mandatory (M), Zeroizable (Z) or Optional (O)

Table 4: Section Data VLP

Order	Element Name	Notes	Section	Format	MZO
1	Section Number X	Coordinates of the Section within the full image	6.4.1.1	ber-oid	M
2	Section Number Y			ber-oid	M
3	Section Range Values	Formatted in ST 1303 [7]	6.4.1.2	MDAP	M
4	Section Uncertainty	Formatted in ST 1303	6.4.1.3	MDAP	Z
5	Plane X-scale Factor	*a* in Equation 8	0	float	O
6	Plane Y-scale Factor	*b* in Equation 8	0	float	O
7	Plane constant value	*c* in Equation 8	0	float	O

6.4.1.1 VLP Element 1 & 2: Section Coordinates

Section Coordinates (i.e., Section Number X and Section Number Y) define the location of where the Section belongs in the full Range Measurement array. Section numbers start with one and always begin in the upper left of the full image and increment towards the lower right. The Section Coordinates are always greater than zero. The Section Coordinates do not exceed the Range Image LS's values for the Number of Sections, i.e., Number of Sections in X and Number of Sections in Y.

With simple Section Layouts, one of the Section Coordinates will always be one. For example, Figure 7 shows five sections, where the bottommost Section has Section Number X equal to 1 and Section Number Y equal to 5.

The Section Number X and Y values use BER-OID encoding (see Motion Imagery Handbook) so they can accommodate any Section size.

Requirement(s)	
ST 1002.3-16	Section Number X shall be greater than or equal to one (1).
ST 1002.3-17	Section Number Y shall be greater than or equal to one (1).
ST 1002.3-18	Section Number X shall be less than or equal to the Range Image LS's Number of Sections in X (Item 17).
ST 1002.3-19	Section Number Y shall be less than or equal to the Range Image LS's Number of Sections in Y (Item 18).
ST 1002.3-20	Where Section Number X is greater than one (1), Section Number Y shall be one (1).
ST 1002.3-21	Where Section Number Y is greater than one (1), Section Number X shall be one (1).

6.4.1.2 VLP Element 3: Section Range Values

The Section Range Values is a partition of Range Measurements array containing either the original range measurements, or the adjusted range measurements produced from the plane subtraction processing (see Section 7). Systems compute adjusted range measurements using

Equation 9. To reverse the computations, receiving systems use the compression parameters given in Section 6.4.1.4 and Equation 10.

MISB ST 1303 specifies how to format an array. The ST 1303 MDAP parameters for the Data Array are:

MDAP ($Note_A$, Section Data, 2, $Note_B$, $Note_B$, $Note_C$)

$Note_A$: Applications determine the type of "Float" or "IMAPB" at runtime

$Note_B$: The dimension size is section dependent; the system provider specifies the size at runtime

$Note_C$: The Element Size is based on whether the system provider is formatting the data as floating-point values or packing the values using IMAPB. When using floating point values, the Element Size will be either 4 or 8 bytes. When using IMAPB the system provider specifies the appropriate precision for the system.

The Section Range Values have a reserved item to represent the Positive Quiet Not-a-Number (+QNaN) value. MISB ST 1201 [8] describes the integer representation of values with reserved bits. The +QNaN signifies no reliable range information exists at a particular pixel within the range image. The zero values in Equation 4 and Equation 5, come from the +QNAN values in the Range Measurements array.

6.4.1.3 VLP Element 4: Section Uncertainty

Use MISB ST 1303 to format the Section Uncertainty values with the option to include the IMAPB min, max values.

Requirement	
ST 1002.1-13	The number of elements in the Section Data VLP – Section Uncertainty shall match the number of elements in the Section Data.

The ST 1303 MDAP parameters for the Section Uncertainty are:

MDAP($Note_D$, 2, $Note_E$, $Note_E$, $Note_F$)

$Note_D$: Applications determine the type of "Float" or "IMAPB" at runtime

$Note_E$: The dimension size is section dependent; the system provider determines the proper size. This value needs to match the Section Range Value's dimension size in Section 6.4.1.2 - $Note_B$

$Note_F$: The Element Size is based on whether the system provider is formatting the data as floating-point values or packing the values using IMAPB. When using floating point values, the Element Size will be either 4 or 8 bytes. When using IMAPB, the system determines the appropriate IMAP precision and hence the IMAP length.

The Section Uncertainty data has a reserved item to represent the Positive Quiet Not-a-Number (+QNaN) value. MISB ST 1201 describes the integer representation of values with reserved bits. The +QNaN signifies no reliable uncertainty information exists at a particular pixel within the range image.

When a Range Image does not have uncertainty values, the length for the Section Uncertainty element is set to zero and systems omit the MDAP from the VLP. Thus, when parsing a Section Data VLP, a zero length for the Section Uncertainty indicates there is not an uncertainty value (i.e., MDAP), but processing continues with the next item in the Section Data VLP. When the whole Range Image does not have uncertainty values all Section Data VLPs for that Range Image will omit the Section Uncertainty.

Requirement(s)	
ST 1002.1-14	The smallest (Min) value for the Section Uncertainly data shall be zero (0).
ST 1002.1-15	If a Range Image does not have uncertainty values, then the Section Uncertainty element's length shall be set to zero (0).

6.4.1.4 VLP Element 5, 6, & 7: Compression Parameters

The optional compression parameters are values for performing Plane Subtraction (see Section 7) on the Section Range Values. Plane Subtraction reduces the dynamic range of the raw range values, thereby reducing the number of bytes needed to represent each value. Plane Subtraction only applies to the Section Range Values, not the Section Uncertainty values. Systems perform Plane Subtraction by computing a reference plane based on a two-dimensional least-squares estimate of a section's original range data, then subtracting the reference plane from the section's original range values producing the Section Range Values. This reduces the large dynamic range of the values for further processing by the Floating Point to Integer Mapping (MISB ST 1201) (referenced in MISB ST 1303). Section 7 fully describes Plane Subtraction. Three compression parameters are specified: Plane X-scale (*a* in Equation 8), Plane Y-scale Factor (*b* in Equation 8), and Plane constant value (*c* in Equation 8). Each of these values is either a 32-bit or 64-bit IEEE floating point value.

6.4.2 Item 17 & 18: Number of Sections in X and Y

Systems may partition the Range Image into one or more Sections. The Number of Sections in X is the count of Sections in the X direction, and the Number of Sections in Y is the count of Sections in the Y direction. Each value is an unsigned integer, represented in BER-OID sub-identifier format (Note: The BER-OID format for these two items is by cause of historical precedent). If the LS does not include the Number of Sections in X or Number of Sections in Y, then the values are default to one (1).

For this version of the standard, only the simple Section layout is prescribed, so either the Number of Sections in X or the Number of Sections in Y must be one as shown in Figure 7 and Figure 8. For example, in Figure 7, the Number of Sections in X equals one and the Number of Sections in Y equals five. In Figure 8, the Number of Sections in X equals three and the Number of Sections in Y equals 1.

Requirement(s)	
ST 1002.3-22	Where Number of Sections X is greater than 1, Number of Sections Y shall be one (1).
ST 1002.3-23	Where Number of Sections Y is greater than 1, Number of Sections X shall be one (1).

6.4.3 Item 19: Generalized Transformation

The Generalized Transformation LS (i.e., MISB ST 1202) specifies a mathematical transformation for projecting information, e.g., points, from one image plane into a second image plane. The Generalized Transformation LS is for specific use cases needing to align a Range Image to a Collaborative Sensor's Image. In this case, the Range Image is a child of the parent Collaborative Image, so use the Child-Parent Transformation (CPT) enumeration 2 defined in Table 1 of ST 1202.

The co-boresighted imaging case is a hardware solution, where multiple focal plane arrays are simultaneously imaging through a single aperture, see Section 5.2. The hardware limits the variation in the perspective centers and the principal axis of the system to be coincident. Therefore, the three-dimensional scene images onto two, dependent, focal plane arrays, where the dependency is in the geometry. These focal planes do not necessarily have the same orientation or identical pixel sizes. Also, the magnification of the two optical paths can be different causing different image scales. The Generalized Transformation LS can sufficiency model these effects. The Value portion of the ST 1202 LS directly inserts into the Range Image LS without any changes (i.e., do not include the ST 1202 key and length in Item 19's value).

7 Plane Subtraction Compression

Range Images have a unique property allowing reductions in the number of bytes when transmitting or storing the data. Typically, the surfaces of a scene are regular, so the Range Image data is relatively uniform (especially in a local area) and smooth (unlike an optical image). For example, an optical image may have textures where adjacent pixels may have a large variation in value (i.e., a white pixel next to a black pixel next to a red pixel, etc.). With Range Images the adjacent pixels are relatively close in value (i.e., 120 meters, next to 125 meters, next to 115 meters, etc.)

Figure 11 illustrates a Range Image of the ground from an airborne system.

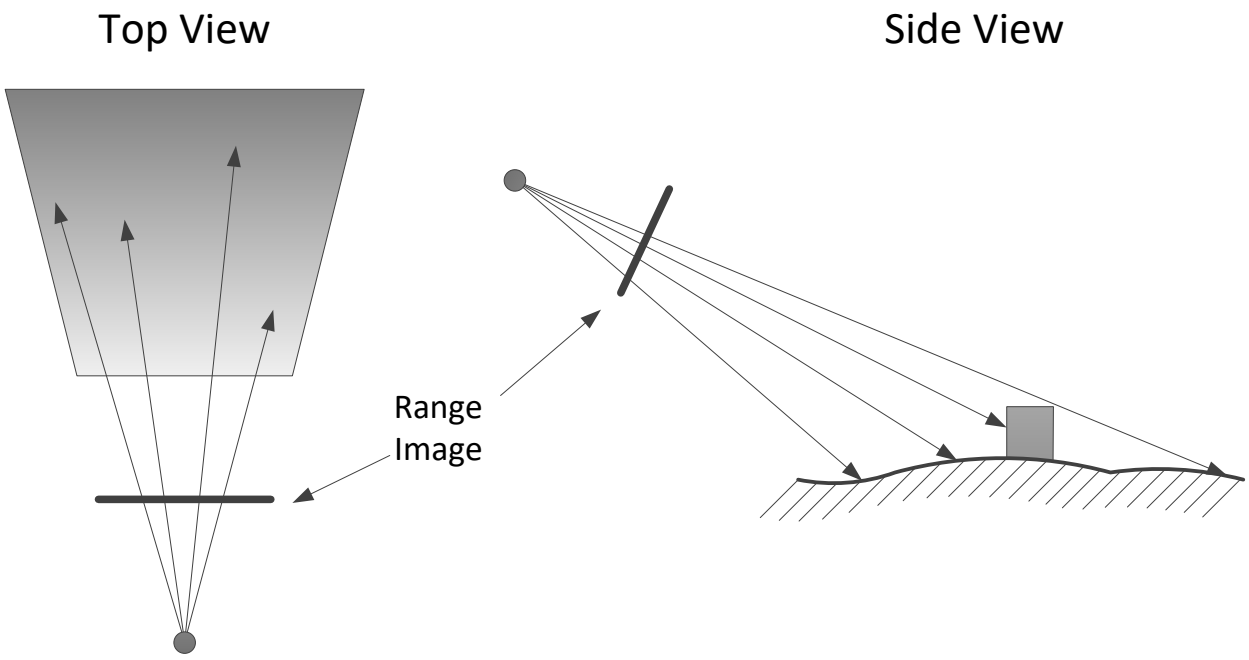


Figure 11: Illustration of Airborne Range Image

Because of this relative uniformity, Range Measurements when fitted to a mathematical plane and subtracted from the original Range Image, produce a residual image with smaller numerical range. The individual values require less bytes providing a level of compression over the original Range Image.

Figure 12 and Figure 13 illustrate a one-dimensional example of the data "compression". Figure 12 shows a plot of 1024 Range Measurements (blue lines). Three of the range measurements (red, purple, and green lines) show the relationship between the physical Range values and the graph. In this example, assume that the Range value is accurate to one meter and the number of bytes to store each range value is based on the dynamic range of the measurements, which in this case is 545 meters. The value of 545 meters requires two bytes (using integers). The total number of bytes for this set of range values is, therefore, 1024 measures x 2 bytes per measure = 2048 bytes.

By fitting a line to this data set (using a least squares estimate), as shown in Figure 13, and then subtracting each range point from the line estimate, the dynamic range reduces to 76 meters, which only requires one byte per measurement.

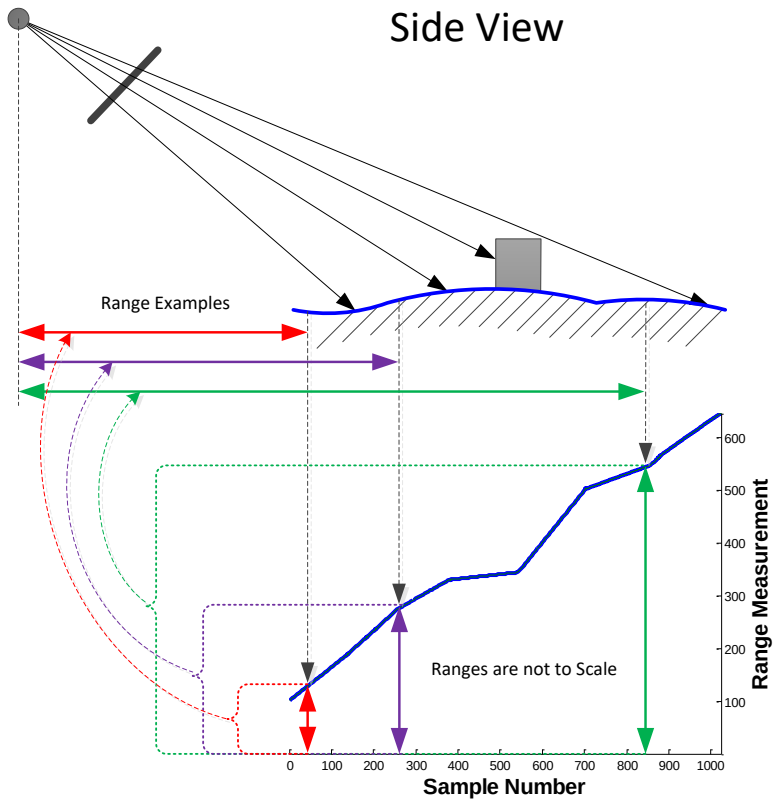


Figure 12: One-Dimensional Range Measurements Illustration

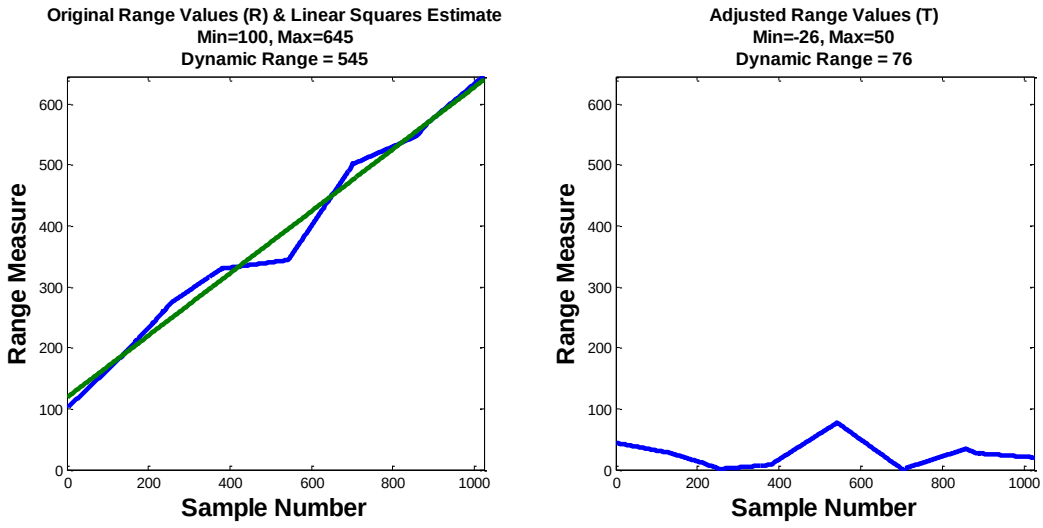


Figure 13: Illustration of Dynamic Range Reduction

(The values in the graph on the right (Adjusted Range Values) increase by the Minimum (-26), so all the values on the blue line would be visible in a graph with the same dynamic range as the original data on the left.)

The fitted line information must accompany the adjusted-valued data to reconstruct the original data; this information requires 4 bytes (2 bytes for the slope and 2 bytes for the y-intercept of a line equation of the form y = ax + b). The total number of bytes for the adjusted range values is: 1024 measures x 1 byte per measure + 2 * 2 bytes = 1028 bytes, which reduces the overall number of bytes by nearly 50%.

The greatest benefit is when the Range Measurements demonstrate a gradual slope with minimal perturbing features. Airborne examples of gradually sloping terrain are: (1) flat, level ground viewed from an oblique angle; and (2) a hill or mountain side.

The following steps show the mathematical algorithm for this example:

1) Let R be the Range Measurements r_i. F is the location, i, of each r_i in the series along with a constant value of one.

$$F = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ \vdots & \vdots \\ N & 1 \end{bmatrix} \quad \text{Equation 1}$$

2) Compute the linear least squares fit, generating the a, b coefficients for the $y = a * i + b$ linear estimate equation.

$$\begin{bmatrix} a \\ b \end{bmatrix} = (F^T * F)^{-1} * F^T * R \quad \text{Equation 2}$$

3) For each Range Measurement subtract the linear estimate.
Let P be the adjusted Range Measurements, p_i.

$$p_i = r_i - (a * i + b) \quad \text{Equation 3}$$

7.1 Plane Subtraction

Plane subtraction is an extension of the previous one-dimensional case to two dimensions. Furthermore, the plane subtraction needs to accommodate “unknown” value processing because with actual range data it is possible to have “unknown” values in the Range Measurements. To compensate for missing data, creating a data mask, M, “zero out’s” and essentially eliminates these values from the computations. The mathematical algorithm uses the following steps:

1) Compute Mask M from Range Measurements

$$M = \begin{bmatrix} m_{1,1} \\ m_{2,1} \\ m_{3,1} \\ \vdots \\ m_{N,1} \\ m_{1,2} \\ m_{2,2} \\ \vdots \\ m_{N,2} \\ m_{1,3} \\ \vdots \\ m_{N,M} \end{bmatrix} \qquad \text{Equation 4}$$

$$\text{Where } m_{i,j} = \begin{cases} 1 & if\ r_{i,j}\ is\ a\ valid\ value \\ 0 & if\ r_{i,j}\ is\ a\ NaN\ value \end{cases} \qquad \text{Equation 5}$$

2) Let R be the Range Measurements of a two-dimensional NxM image with each element denoted as $r_{i,j}$. Form $\vec{R}$ from R by vectorizing the image into one long vector and applying the mask. Note: $\circledast$ is point-wise multiplication.

$$\vec{R} = \begin{bmatrix} r_{1,1} \\ r_{2,1} \\ r_{3,1} \\ \vdots \\ r_{N,1} \\ r_{1,2} \\ r_{2,2} \\ \vdots \\ r_{N,2} \\ r_{1,3} \\ \vdots \\ r_{N,M} \end{bmatrix} \circledast M \qquad \text{Equation 6}$$

3) F is the location, *i,j* of each $r_{i,j}$ in the series along with a constant value of one. Note: $\circledast$ is point-wise multiplication.

$$F = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \\ \vdots & \vdots & \vdots \\ N & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 2 & 1 \\ \vdots & \vdots & \vdots \\ N & 2 & 1 \\ 1 & 3 & 1 \\ \vdots & \vdots & \vdots \\ N & M & 1 \end{bmatrix} \circledast [M \quad M \quad M] \qquad \text{Equation 7}$$

4) Compute the planer least squares fit using $y = a * i + b * j + c$ to compute the *a*, *b* and *c* coefficients for the planer estimate equation

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = (F^T * F)^{-1} * F^T * \vec{R} \quad \text{Equation 8}$$

5) For each Range Measurement subtract the planer estimate. Let P be the Adjusted Range Measurements with each element of the image denoted as $p_{i,j}$.

$$p_{i,j} = r_{i,j} - (a * i + b * j + c) \quad \text{Equation 9}$$

Normally, it is not recommended to use least squares to produce the coefficients because of poor conditioning[1] of the matrix in Equation 8; however, in this situation the results only produce a reference plane estimate used in reverse (Equation 10) for the decompressing and obtaining the original Range Measurements. In this case, the complete accuracy of the estimate is not a concern with this type of data. The least squares method is employed here because of its simplicity; other methods, such as QR factorization or SVD, can achieve better estimates of the coefficients with a cost in complexity and computation time.

7.2 Reverse Plane Subtraction

To reverse the plane subtraction processing use the following equation:

$$r_{i,j} = p_{i,j} + (a * i + b * j + c) \quad \text{Equation 10}$$

Store the parameters *a*, *b*, and *c* with the adjusted range measurements.

8 Retired Requirements

Requirement(s)	
ST 1002.1-01 (Retired)	The Range Image Local Set shall have the UL Key 06.0E.2B.34.02.0B.01.01.0E.01.03.03.0C.00.00.00 (CRC 41152).
ST 1002.1-10 (Retired)	Section Data Variable Length Packs shall always contain Section Number X and Section Number Y.
ST 1002.1-11 (Retired)	One of the Section Coordinate Values, either Section Number X or Section Number Y, shall be one (1).
ST 1002.1-12 (Retired)	Where reassembling a full image, all Sections shall be assembled in order from the lowest value X/Y Section Coordinates to the largest X/Y Section Coordinates.
ST 1002.1-04 (Retired)	A Range Image Local Set shall include the CRC-16-CCITT (Item 21) in accordance with Motion Imagery Handbook, "Local Set Check Value Implementation" instructions.

[1] See advanced Linear Algebra textbooks for discussion on Condition Numbers of matrices.